The Stark-Zeeman Line-Shape Code

User Manual and Theoretical Guide

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Open-Source Coupled Stark-Zeeman Plasma Line-Shape Project

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1 Introduction

Modeling the spectral line shapes of multi-electron ions in hot, dense plasmas is a challenging problem in plasma physics, astrophysics, and laboratory nuclear fusion diagnostics. The presence of strong external magnetic fields ($\vec{B}$) coupled with the stochastic electric microfields ($\vec{F}$) produced by moving ions and electrons in the plasma leads to a complex, simultaneous splitting and broadening known as the Stark-Zeeman (SZ) effect.

The **Stark Zeeman Line-Shape** (**StarkZee**) code is an open-source Python implementation that couples atomic physics calculations in arbitrary magnetic fields with plasma line-broadening theories under quasi-static electric microfields and electron impact. The library incorporates ion dynamics and electric field fluctuations via the Frequency Fluctuation Model (FFM).

This manual details the theoretical background, governing equations, code structure, physical simplifications, and high-performance NumPy vectorization optimizations implemented in the package.

2 Theoretical Background and Governing Equations

2.1 Liouville Space Representation

The spectral line-shape intensity $I(\omega)$ of emitted radiation is related to the Fourier transform of the dipole autocorrelation function $C(t)$ by [2]:

$$I(\omega) = \frac{1}{\pi} \operatorname{Re} \int_0^\infty C(t)\, e^{i\omega t}\, dt \tag{1}$$

In Liouville space notation, the autocorrelation function is a trace over the quantum emitter states:

$$C(t) = \langle\langle \vec{d}^{\,*} | U(t) | \vec{d} \rho_0 \rangle\rangle \tag{2}$$

where $\vec{d}$ is the electric transition dipole operator, ρ_0 is the equilibrium density matrix of the initial manifold, and $U(t)$ is the time-evolution propagator.

In the anti-symmetric subspace (ground-to-excited manifold transitions in the no-quenching approximation), the Liouvillian operator L is constructed from the excited-state Hamiltonian H_u and ground-state Hamiltonian H_l:

$$L = \frac{1}{\hbar}\left(H_u \otimes \hat{I}^d - \hat{I} \otimes H_l^d \right) \tag{3}$$

The diagonal elements of L give the field-free transition frequencies $\omega_{ij} = (E_i^u - E_j^l)/\hbar$; off-diagonal elements encode the Stark and Zeeman couplings between dressed states.

From propagator to resolvent. The time-evolution propagator is generated by L:

$$U(t) = e^{-iLt} \tag{4}$$

Substituting into Eq. (2) gives the explicit correlation function

$$C(t) = \langle\langle \vec{d}^{\,*} | e^{-iLt} | \vec{d} \rho_0 \rangle\rangle \tag{5}$$

Inserting this into Eq. (1) and evaluating the one-sided Fourier transform converts the time-domain exponential into a frequency-domain resolvent:

$$I(\omega) = \frac{1}{\pi} \mathrm{Re} \, \langle\langle \vec{d}^* | \, (\omega \, \hat{I} - L)^{-1} \, | \vec{d} \rho_0 \rangle\rangle \tag{6}$$

Fast electron collisions are encoded as an additional imaginary damping operator $\Phi(\omega)$ in the same Liouville space (Section 2.8), shifting $L \to L + i\Phi(\omega)$:

$$I(\omega) = \frac{1}{\pi} \mathrm{Re} \, \langle\langle \vec{d}^* | \, \big(\omega \, \hat{I} - L - i \, \Phi(\omega)\big)^{-1} \, | \vec{d} \rho_0 \rangle\rangle \tag{7}$$

Field-dependent Liouvillian and static profile. Under the quasi-static ion approximation the radiator is exposed to a static ionic microfield $\vec{F}$ at angle θ to $\vec{B}$. Both manifold Hamiltonians H_u and H_l then depend on the field magnitude F and orientation $\mu = \cos\theta$ through the Stark perturbation V_E (Section 2.4), making the Liouvillian parametrically field-dependent:

$$L(F,\mu) = \frac{1}{\hbar}\Big[H_u(F,\mu) \otimes \hat{I}^d - \hat{I} \otimes H_l^d(F,\mu)\Big], \qquad H(F,\mu) = H_A + V_E(F,\mu) \tag{8}$$

where H_A is the field-free magnetic Hamiltonian (Section 2.2) and $V_E(F,\mu)$ is the linear Stark interaction (Section 2.4). Equations (7) and (8) together give the field-dependent q-polarized profile at a given configuration (F,μ):

$$I_q(\omega,F,\mu) = \frac{1}{\pi} \mathrm{Re} \, \langle\langle d_q^* \, | \, \big(\omega \, \hat{I} - L(F,\mu) - i \, \Phi(\omega)\big)^{-1} \, | \, d_q \, \rho_0 \rangle\rangle \tag{9}$$

The total q-polarized profile is then the average over the ionic microfield distribution $W(F)$ and orientation:

$$I_q(\omega) = \frac{1}{2} \int_0^\infty \int_{-1}^1 W(F) \, I_q(\omega,F,\mu) \, d\mu \, dF \tag{10}$$

Because $I_q(\omega,F,\mu) = I_q(\omega,F,-\mu)$ (the Hamiltonian is even in μ via $F_z = F\mu$ and $F_x = F\sqrt{1-\mu^2}$), the integral over $[-1,1]$ reduces to twice the $[0,1]$ range, discretized by Gauss–Legendre quadrature. The observed intensity at angle α between the line of sight and $\vec{B}$ is:

$$I(\omega,\alpha) = I_\pi(\omega)\sin^2\!\alpha + \tfrac{1}{2}\big(I_{\sigma^+}(\omega) + I_{\sigma^-}(\omega)\big)(1+\cos^2\!\alpha) \tag{11}$$

Sections 2.2–2.8 detail each ingredient of Eq. (9); Section 2.11 extends this framework to include ion dynamics via the FFM.

2.2 Radiator Hamiltonian in Magnetic Fields

For each principal quantum number shell n, the atomic Hamiltonian H_A is constructed in the canonical uncoupled hydrogenic basis $|n, l, m_l, s{=}\tfrac{1}{2}, m_s\rangle$ of dimension $2n^2$. Basis states are ordered by an outer loop over $l \in [0, n-1]$, an intermediate loop over $m_l \in [-l, l]$, and an inner loop over $m_s \in \{+\tfrac{1}{2}, -\tfrac{1}{2}\}$. Five physical contributions are assembled into a single matrix [3]:

$$H_A = H_0 + H_{\mathrm{SO}} + H_{\mathrm{FS}} + H_Z^{(1)} + H_Z^{(2)} \tag{12}$$

2.2.1 Unperturbed Hydrogenic Energy (H_0)

The diagonal unperturbed energy is corrected for the finite nuclear mass:

$$\langle n, l, m_l, m_s | H_0 | n, l, m_l, m_s \rangle = -\frac{Z^2 \,\mathrm{Ry}_{\mathrm{red}}}{n^2}, \qquad \mathrm{Ry}_{\mathrm{red}} = \mathrm{Ry}_\infty \frac{M_{\mathrm{nuc}}}{m_e + M_{\mathrm{nuc}}} \tag{13}$$

with A_{emitter} the atomic mass number of the emitting isotope. When `use_empirical_data=True` this analytic diagonal is replaced by NIST tabulated energies (Section 2.3).

2.2.2 Spin-Orbit Coupling (H_{SO})

The spin-orbit interaction $H_{\mathrm{SO}} = \xi_{nl} \vec{L} \cdot \vec{S}$ couples orbital and spin angular momenta. The coupling constant for $l > 0$ is:

$$\xi_{nl} = \frac{Z^4 \alpha^2 \,\mathrm{Ry}_\infty}{n^3 \, l \,(l + \frac{1}{2})\,(l+1)} \tag{14}$$

Using the ladder decomposition $\vec{L} \cdot \vec{S} = L_z S_z + \frac{1}{2}(L_+ S_- + L_- S_+)$, the matrix elements within a given l-subshell are:

- **Diagonal** ($\Delta m_l = 0$, $\Delta m_s = 0$):

$$\langle l, m_l, m_s | H_{\mathrm{SO}} | l, m_l, m_s \rangle = \xi_{nl}\, m_l\, m_s \tag{15}$$

- **Off-diagonal** ($\Delta m_l = \pm 1$, $\Delta m_s = \mp 1$): for $s = \frac{1}{2}$ the spin ladder factor equals 1 for all allowed transitions:

$$\langle l, m_l \pm 1, m_s \mp 1 | H_{\mathrm{SO}} | l, m_l, m_s \rangle = \tfrac{1}{2}\, \xi_{nl} \sqrt{l(l+1) - m_l(m_l \pm 1)} \tag{16}$$

2.2.3 Fine-Structure Relativistic Corrections (H_{FS})

To restore the Dirac degeneracy (e.g. $2s_{1/2}$ and $2p_{1/2}$), the mass-velocity and Darwin terms are added [3]. These are diagonal in $|n, l, m_l, m_s\rangle$ and independent of m_l and m_s:

$$\langle l | H_{\mathrm{FS}} | l \rangle = \begin{cases} -A_{\mathrm{FS}}\big(n - \frac{3}{4}\big) & l = 0 \\ -A_{\mathrm{FS}}\left(\dfrac{n}{l + \frac{1}{2}} - \dfrac{3}{4}\right) & l > 0 \end{cases} \tag{17}$$

where $A_{\mathrm{FS}} = Z^4 \alpha^2 \,\mathrm{Ry}_\infty / n^4$. The algebraic identity $H_{\mathrm{SO}} + H_{\mathrm{FS}}$ reproduces the exact Dirac fine-structure eigenvalues for both $j = l \pm \frac{1}{2}$.

2.2.4 Linear Zeeman Effect ($H_Z^{(1)}$)

$$\langle m_l, m_s | H_Z^{(1)} | m_l, m_s \rangle = \mu_B B\,(m_l + g_s m_s) \tag{18}$$

where $\mu_B \approx 5.788 \times 10^{-5}$ eV/T and $g_s \approx 2.0023192$.

2.2.5 Quadratic (Diamagnetic) Zeeman Effect ($H_Z^{(2)}$)

$$H_Z^{(2)} = \frac{e^2 B^2}{8 m_e}\, r^2 \sin^2\theta \tag{19}$$

This operator couples states with $\Delta l = 0$ and $\Delta l = \pm 2$. The matrix elements factor into radial and angular parts:

$$\langle l_1, m_l | H_Z^{(2)} | l_2, m_l \rangle = \left(\frac{e^2 B^2 a_0^2}{8 m_e}\right) \langle n, l_1 | r^2 | n, l_2 \rangle \; \langle l_1, m_l | \sin^2\theta | l_2, m_l \rangle \tag{20}$$

1. **Radial integrals**:
   - Diagonal ($l_1 = l_2 = l$): $\langle n,l|r^2|n,l\rangle = \dfrac{n^2}{2Z^2}\left[5n^2 + 1 - 3l(l+1)\right]\ [a_0^2]$
   - Off-diagonal ($l_2 = l_1 \pm 2$): evaluated numerically via $\int_0^\infty R_{nl_1}\, r^4\, R_{nl_2}\, dr$, cached with `lru_cache`.
2. **Angular integrals**:
   - Diagonal: for $l = 0$, $\langle \sin^2\theta\rangle = 2/3$; for $l > 0$,

$$\langle l, m_l|\sin^2\theta|l, m_l\rangle = 1 - \frac{2l^2 + 2l - 1 - 2m_l^2}{(2l-1)(2l+3)} \tag{21}$$

   - Off-diagonal ($l_2 = l_1 + 2$, $l_{\text{low}} = \min(l_1, l_2)$): the constant term in $1 - \cos^2\theta$ vanishes for orthogonal harmonics, leaving:

$$\langle l_1, m_l|\sin^2\theta|l_2, m_l\rangle = -\sqrt{\frac{\left[(l_{\text{low}}+1)^2 - m_l^2\right]\left[(l_{\text{low}}+2)^2 - m_l^2\right]}{(2l_{\text{low}}+1)(2l_{\text{low}}+3)^2(2l_{\text{low}}+5)}} \tag{22}$$

2.3 Empirical Field-Free Energies

The analytic diagonal $H_0 = -Z^2\,\text{Ry}_{\text{red}}/n^2$, together with the spin-orbit and fine-structure terms, reproduces the field-free level positions only to the accuracy of the hydrogenic Dirac formula. For quantitative line centers, `StarkZee` can instead inject *measured* field-free energies through the `use_empirical_data=True` flag (with the element symbol `atom`). The values are read from a tabulated database (`atomic_data.load_levels`) holding NIST level energies in wavenumbers [cm^{-1}], resolved by (n, l, j). The empirical Hamiltonian is kept in cm^{-1} throughout this code path so that callers can work entirely in wavenumber units; all Zeeman contributions are also converted from eV to cm^{-1} before being added, keeping every term on the same scale.

Because the field-free Hamiltonian is degenerate (the Dirac formula makes $2s_{1/2}$ and $2p_{1/2}$ coincide), the empirical energies cannot simply be written on the diagonal of the uncoupled $|n, l, m_l, m_s\rangle$ basis: `numpy.linalg.eigh` is free to mix the degenerate l states arbitrarily. `StarkZee` therefore:

1. diagonalizes a degeneracy-broken field-free Hamiltonian (unperturbed energy + spin-orbit only, *omitting* the mass-velocity/Darwin term so that l remains a good label), yielding eigenvectors V;
2. labels each coupled eigenstate k by its dominant orbital component l and its total angular momentum $j = l \pm \frac{1}{2}$, the branch being set by the sign of $\langle \vec{L}\cdot\vec{S}\rangle = (E_k - E_n)/\xi_{nl}$;
3. assigns the tabulated energy $D_k = E^{\text{emp}}(l, j)$ [cm^{-1}] to each eigenstate and reconstructs the field-free Hamiltonian as

$$H_0^{\text{emp}} = V\,\text{diag}(D)\,V^\dagger \quad [\text{cm}^{-1}]. \tag{23}$$

The Zeeman terms ($\mu_B B(m_l + g_s m_s)$ and the diamagnetic correction) are converted from eV to cm^{-1} and added on top of H_0^{emp}. Since the tabulated values are absolute level energies (ground state at 0), transition wavenumbers follow directly as differences $E_u - E_l$ and reproduce the observed NIST Lyman/Balmer line centers.

2.4 Full Electron-Radiator Hamiltonian

Under the quasi-static ion approximation, the radiator is subjected to a constant electric microfield $\vec{F}$ at an angle θ relative to the magnetic field $\vec{B}$. The microfield is decomposed into longitudinal ($F_z = F\cos\theta$) and transverse ($F_x = F\sin\theta$) components. The Stark interaction is

$$V_E = -e\,\vec{r}\cdot\vec{F} = -e(zF_z + xF_x) = -\hat{\boldsymbol{d}}\cdot\vec{F} \tag{24}$$

where $\hat{\boldsymbol{d}} = e\vec{r}$ is the electric dipole operator. The full electron-radiator Hamiltonian is

$$H = H_A + V_E = H_0 + V_{\mathrm{SO}} + H_Z^{(1)} + H_Z^{(2)} + V_E \tag{25}$$

H_A and V_E are assembled into a single matrix and diagonalized *as a whole* by `numpy.linalg.eigh` at every microfield quadrature point (`solve_starkzee`). The Stark interaction is **not** treated perturbatively: there is no expansion in powers of V_E. This exact treatment is essential when the Stark energy $eF\langle r\rangle_n$ is comparable to the fine-structure or Zeeman splittings, yielding the Stark-dressed eigenstates $|k\rangle$ and transition frequencies ω_k.

2.4.1 Stark Perturbation Matrix Elements

The selection rule for the within-shell Stark coupling is $\Delta n = 0$, $|l_i - l_j| = 1$, $\Delta m_s = 0$. In energy units [eV]:

$$\langle n, l_i, m_{l,i}|V_E|n, l_j, m_{l,j}\rangle = -\,\delta_{m_{s,i},m_{s,j}}\,\langle n, l_i|r|n, l_j\rangle\,\big\langle l_i, m_{l,i}\big|(zF_z + xF_x)/r\big|l_j, m_{l,j}\big\rangle \tag{26}$$

Radial matrix element. For $l = \max(l_i, l_j)$, Gordon's formula gives the within-shell radial dipole element exactly:

$$\langle n, l|r|n, l-1\rangle = \frac{3n}{2Z}\sqrt{n^2 - l^2} \quad [a_0] \tag{27}$$

Angular matrix elements. Writing $z/r = T_0^{(1)}$ and $x/r = (T_{-1}^{(1)} + T_{+1}^{(1)})/\sqrt{2}$:

- $q = 0$ ($\Delta m_l = 0$): $\langle l, m_l|\cos\theta|l+1, m_l\rangle = \sqrt{\dfrac{(l+1)^2 - m_l^2}{(2l+1)(2l+3)}}$
- $q = +1$ ($\Delta m_l = -1$):

$$\langle l_1, m_{l,1}|T_{+1}^{(1)}|l_2, m_{l,2}\rangle = \begin{cases} +\sqrt{\dfrac{(l-m_{l,2})(l-m_{l,2}-1)}{2(2l-1)(2l+1)}} & l_1 = l_2 - 1 \\ -\sqrt{\dfrac{(l+m_{l,2})(l+m_{l,2}+1)}{2(2l-1)(2l+1)}} & l_1 = l_2 + 1 \end{cases} \tag{28}$$

- $q = -1$ ($\Delta m_l = +1$):

$$\langle l_1, m_{l,1}|T_{-1}^{(1)}|l_2, m_{l,2}\rangle = \begin{cases} -\sqrt{\dfrac{(l+m_{l,2})(l+m_{l,2}-1)}{2(2l-1)(2l+1)}} & l_1 = l_2 - 1 \\ +\sqrt{\dfrac{(l-m_{l,2})(l-m_{l,2}+1)}{2(2l-1)(2l+1)}} & l_1 = l_2 + 1 \end{cases} \tag{29}$$

where $l = \max(l_1, l_2)$ in all cases. The Stark matrices M_z and M_x [eV/(V/m)] are pre-computed once per (n, Z) and cached (`_stark_templates`), so that $V_E = F_z M_z + F_x M_x$ is formed at each quadrature point by a simple scalar multiplication.

2.5 Transition Dipole Operator

The transition dipole matrices D_q ($q \in \{0, \pm 1\}$) are built in the uncoupled basis $|n, l, m_l, s, m_s\rangle$. Selection rules require $\Delta l = \pm 1$, $\Delta m_l = q$, and $\Delta m_s = 0$:

$$\langle n_l, l_l, m_{l,l}, m_{s,l}|D_q|n_u, l_u, m_{l,u}, m_{s,u}\rangle = \delta_{m_{s,l},m_{s,u}} \langle n_l, l_l|r|n_u, l_u\rangle \langle l_l, m_{l,l}|T_q^{(1)}|l_u, m_{l,u}\rangle \tag{30}$$

2.5.1 Angular Dipole Matrix Elements

The angular factors $\langle l_l, m_{l,l}|T_q^{(1)}|l_u, m_{l,u}\rangle$ follow from the Wigner-Eckart theorem [5]:

- For $q = 0$ ($\Delta m_l = 0$):

$$\langle l, m_l|T_0^{(1)}|l+1, m_l\rangle = \sqrt{\frac{(l+1)^2 - m_l^2}{(2l+1)(2l+3)}} \tag{31}$$

- For $q = +1$ ($\Delta m_l = -1$):

$$\langle l_1, m_{l,1}|T_{+1}^{(1)}|l_2, m_{l,2}\rangle = \begin{cases} +\sqrt{\dfrac{(l-m_{l,2})(l-m_{l,2}-1)}{2(2l-1)(2l+1)}} & l_1 = l_2 - 1 \\ -\sqrt{\dfrac{(l+m_{l,2})(l+m_{l,2}+1)}{2(2l-1)(2l+1)}} & l_1 = l_2 + 1 \end{cases} \tag{32}$$

- For $q = -1$ ($\Delta m_l = +1$):

$$\langle l_1, m_{l,1}|T_{-1}^{(1)}|l_2, m_{l,2}\rangle = \begin{cases} -\sqrt{\dfrac{(l+m_{l,2})(l+m_{l,2}-1)}{2(2l-1)(2l+1)}} & l_1 = l_2 - 1 \\ +\sqrt{\dfrac{(l-m_{l,2})(l-m_{l,2}+1)}{2(2l-1)(2l+1)}} & l_1 = l_2 + 1 \end{cases} \tag{33}$$

where $l = \max(l_1, l_2)$.

2.5.2 Hydrogenic Radial Dipole Integrals (Gordon's Formula)

For $n_u \neq n_l$, the inter-shell radial elements $\langle n_l, l_l|r|n_u, l_u\rangle$ are evaluated exactly using Gordon's (1929) analytical formula [4]:

$$\begin{aligned}\langle n_l, l_l|r|n_u, l_u\rangle = \frac{1}{Z}\,\frac{(-1)^{n_l-L}}{4(2L-1)!}\sqrt{\frac{(n_u+L)!\,(n_l+L-1)!}{(n_u-L-1)!\,(n_l-L)!}} \\ \times \frac{(4n_u n_l)^{L+1}(n_u-n_l)^{n_u+n_l-2L-2}}{(n_u+n_l)^{n_u+n_l}}\left[F_1 - \left(\frac{n_u-n_l}{n_u+n_l}\right)^2 F_2\right]\end{aligned} \tag{34}$$

where $L = \max(l_u, l_l)$, and F_1, F_2 are terminating Gauss hypergeometric functions:

$$F_1 = {}_2F_1\left(-n_u + L + 1,\ -n_l + L,\ 2L,\ -\frac{4n_u n_l}{(n_u-n_l)^2}\right) \tag{35}$$

$$F_2 = {}_2F_1\left(-n_u + L - 1,\ -n_l + L,\ 2L,\ -\frac{4n_u n_l}{(n_u-n_l)^2}\right) \tag{36}$$

Because the first argument in both hypergeometric functions is a non-positive integer, each series terminates as a finite sum:

$${}_2F_1(-m, b, c, x) = \sum_{k=0}^{m} \frac{(-m)_k\,(b)_k}{(c)_k\,k!}\,x^k \tag{37}$$

where $(-m)_k$ is the Pochhammer symbol. This exact evaluation carries no integration cutoff error and is cached for reuse.

2.6 Diagonalization and Transition Dipole Basis Rotation

At each quadrature point (F, μ) the total Hamiltonians for both manifolds are:

$$H_u(F,\mu) = H_{A,u} + V_{E,u}(F_z, F_x), \qquad H_l(F,\mu) = H_{A,l} + V_{E,l}(F_z, F_x) \tag{38}$$

Diagonalization via `numpy.linalg.eigh` gives eigenvectors V_u, V_l and eigenvalues E_u^p, E_l^k:

$$V_u^\dagger H_u V_u = \mathrm{diag}(E_u^p), \qquad V_l^\dagger H_l V_l = \mathrm{diag}(E_l^k) \tag{39}$$

The transition energy for each dressed-state pair $(p \to k)$ is $E_{pk} = E_u^p - E_l^k$. The uncoupled dipole matrices D_q are rotated into the mixed eigenstate bases:

$$D_q' = V_l^\dagger\, D_q\, V_u, \qquad S_q(p,k) = |D_q'(k,p)|^2 \tag{40}$$

The weight $S_q(p,k)$ and energy E_{pk} together define one Stark-Dressed Transition (SDT). The static profile at field configuration (F, μ) — Eq. (9) — is then realized in practice as a sum over all SDTs weighted by the quadrature weight W_{ij}:

$$I_q(\omega, F, \mu) = \sum_{p,k} W_{ij}\, S_q(p,k)\, \frac{\gamma_e/\pi}{(\omega - E_{pk})^2 + \gamma_e^2} \tag{41}$$

where γ_e is the electron-impact half-width of Section 2.8.

2.7 Plasma Microfield Distributions

The electric microfield is averaged over a probability distribution $W(F)$. The presence of $\vec{B}$ breaks spherical symmetry, so the integration must be performed over both the field magnitude $F = |\vec{F}|$ and the orientation $\mu = \cos\theta$ between $\vec{F}$ and $\vec{B}$.

2.7.1 The Hooper Screened Microfield Distribution

To account for Debye screening of the ion Coulomb fields by surrounding plasma electrons, StarkZee implements the **Hooper microfield distribution** [11]:

$$W(\beta, a) = \frac{2\beta}{\pi} \int_0^\infty y \sin(\beta y)\, T(y, a)\, dy \tag{42}$$

where:

- $\beta = F/F_0$ is the reduced electric field strength.
- F_0 is the **Holtsmark normal field strength**, the Coulomb field of a single perturber at the mean inter-particle distance r_e:

$$F_0 = \frac{e}{4\pi\varepsilon_0 r_e^2}, \qquad r_e = \left(\frac{3}{4\pi N_e}\right)^{1/3} \tag{43}$$

- $a = r_e/\lambda_D$ is the Debye screening parameter, where λ_D is the multi-species Debye length:

$$\lambda_D = \sqrt{\frac{\varepsilon_0 k_B T_e}{N_e e^2 \left(1 + \sum_i X_i Z_i^2\right)}} \tag{44}$$

  with $X_i = N_i/N_e$ the fractional ion concentration.

- $T(y,a) = \exp\big(-y^{3/2} S(y,a)\big)$ is the screened characteristic function, with Debye screening factor:

$$S(y,a) = \left(1 + \frac{1.5\,a^2}{y^2}\right)^{-3/4} \tag{45}$$

Setting $a = 0$ yields $S = 1$ and $T = \exp(-y^{3/2})$, recovering the unscreened **Holtsmark distribution** $W_H(\beta)$.

2.7.2 The Potekhin Microfield Distribution

The analytical fits of Potekhin et al. (2002) [12] describe the cumulative distribution $Q(\beta)$ and probability density $P(\beta)$ for both screened ($s > 0$) and unscreened ($s = 0$) potentials, for neutral and charged radiators, incorporating the ion-ion coupling parameter Γ:

$$Q(\beta) = \frac{q_0\beta^3 - 1.33\,\beta^{9/2} + \beta^6}{q_1 + q_2\beta^2 + q_3\beta^3 - \frac{1}{3}\beta^{9/2} + \beta^6} \quad (s = 0) \tag{46}$$

$$P(\beta) \approx \beta^2 S_N \left[A\, e^{-a\beta^\alpha} + B\, e^{-b\beta^\gamma} + \frac{e^{-\Gamma\beta^{1/2}}}{1 + c\beta^{9/2}} \right] \quad (s > 0) \tag{47}$$

where $q_n, A, a, \alpha, B, b, \gamma, c$ are tabulated functions of Γ and s, and S_N is the normalization constant.

2.7.3 Quadrature Discretization

The double integral $\iint W(F,\mu)\, dF\, d\mu$ (Eq. 10) is discretized as follows:

1. **Field magnitude** F: A uniform grid of N_F points in reduced field space $\beta_i \in [0, \beta_{\max}]$ (default $\beta_{\max} = 10$):

$$F_i = \beta_i F_0, \qquad W_{F,i} = W(\beta_i, a)\,\Delta\beta, \qquad \sum_i W_{F,i} = 1 \tag{48}$$

2. **Field orientation** μ: N_μ-point **Gauss-Legendre quadrature** over $[0,1]$. Legendre roots $x_j \in [-1,1]$ and weights w_j are mapped as:

$$\mu_j = \frac{x_j + 1}{2}, \qquad W_{\mu,j} = \frac{w_j}{2} \tag{49}$$

3. **Combined weight**: $W_{ij} = W_{F,i}\, W_{\mu,j}$. Points with $W_{ij} < 10^{-15}$ are skipped.

At each active point the field decomposes as:

$$F_{z,ij} = F_i\,\mu_j, \qquad F_{x,ij} = F_i\sqrt{1 - \mu_j^2} \tag{50}$$

1. **Holtsmark Distribution**: Standard unscreened microfield for weakly coupled plasmas [10]:

$$W_H(\beta) = \frac{2\beta}{\pi} \int_0^\infty y \sin(\beta y) e^{-y^{1.5}} dy \tag{51}$$

   where $\beta = F/F_0$ is the normalized field strength.

2. **Hooper Distribution**: Debye-screened microfield distribution [11]:

$$W_{Hooper}(\beta, a) = \frac{2\beta}{\pi} \int_0^\infty y \sin(\beta y) e^{-y^{1.5}\chi(y,a)} dy \tag{52}$$

   where $\chi(y,a) = (1 + 1.5a^2/y^2)^{-0.75}$ is the screening correction fit, and $a = r_e/\lambda_D$ is the ratio of the ion-sphere radius to the Debye length.

3. **Potekhin Distribution**: Analytical fits for electric microfield distributions $P(\beta)$ in plasmas [12], which support both screened and unscreened cases as well as neutral and charged radiator points, incorporating the ion-ion coupling parameter Γ.

2.8 Electron Impact Broadening

Fast-moving electrons are treated in the *impact* (completed-collision) approximation: each collision is completed in a time short compared to both the inverse linewidth and the mean inter-collision time. In this limit, successive collisions are statistically independent and their net effect on the line shape enters through a damping operator $\Phi(\omega)$ in the Liouville-space resolvent (Eq. 7).

2.8.1 *Liouville-space structure of $\Phi(\omega)$ and connection to $\langle r^2 \rangle$*

In the no-quenching approximation (electron collisions do not mix the upper and lower radiating manifolds), the operator Φ in Liouville space factorizes as a sum of independent contributions from each manifold:

$$\Phi(\omega) = \tfrac{1}{2}\big(\Phi_u(\omega) \otimes \hat{I}^d + \hat{I} \otimes \Phi_l^d(\omega)\big) \tag{53}$$

where Φ_u and Φ_l are self-energy operators acting within the upper (n_u) and lower (n_l) manifolds respectively.

The GBK semi-classical model evaluates Φ_n from the leading long-range term of the electron–radiator interaction, which is the dipole–dipole coupling $V_{ee} \propto r^2_{\text{atom}}/r_e^3$. Averaging over the Maxwell–Boltzmann distribution of electron impact parameters and velocities, this yields a self-energy proportional to the *operator* r_n^2 within shell n:

$$\Phi_n(\omega) = W_0(\omega)\, r_n^2 \left[C_n + G(\omega)\right] \tag{54}$$

where r_n^2 is the mean-square-displacement operator within shell n, W_0 is the density-temperature prefactor (below), and $G(\omega)$ is the frequency-dependent GBK factor (below). The r^2 dependence is the direct quantum-mechanical trace of the dipole-coupling cross-section: a state with larger electronic extent couples more strongly to a passing electron.

Rotating to the SDT basis via the eigenvectors of $H_n(F, \mu)$ (Section 2.6) makes Φ_n approximately diagonal, since the dressed-state splitting $\omega_{kk'}$ is generally large compared to the collision rate. The diagonal element for upper dressed state $|k\rangle$ is:

$$\gamma_k^{(u)} = W_0\, \langle k\,|\, r_u^2 \,|\, k\rangle \left[C_{n_u} + G(\Delta E_k)\right] \tag{55}$$

The resolvent $(\omega \hat{I} - L(F, \mu) - i\Phi(\omega))^{-1}$ then has eigenvalues $(\omega - \omega_k - i\gamma_k)^{-1}$ in the SDT basis, and Eq. (9) reduces directly to the Lorentzian sum of Eq. (73).

Shell-average approximation. Computing the per-SDT matrix element $\langle k|r_u^2|k\rangle$ exactly requires rotating the r^2 operator to each dressed-state basis at every quadrature point (`electron_operator=True`). The default approximation replaces $\langle k|r_u^2|k\rangle$ by the shell average $\langle r^2 \rangle_{n_u}$, giving a single homogeneous half-width:

$$\gamma_e(\Delta E) = W_0\, \langle r^2 \rangle_{n_u} \left[C_{n_u} + G(\Delta E)\right] \tag{56}$$

The lower-manifold contribution $\gamma_k^{(l)}$ is retained in principle through Eq. (53); in practice, for Balmer lines ($n_l = 2$, $\langle r^2 \rangle_2 = 12\, a_0^2$) it is a factor $\sim$3 smaller than the $n_u = 3$ upper term and is dominated by the latter.

2.8.2 Density-Temperature Prefactor

$$W_0 = \frac{4\pi}{3} N_e \sqrt{\frac{2m_e}{\pi k_B T_e}} \left(\frac{\hbar}{m_e}\right)^2 \frac{\hbar}{e} \quad [\mathrm{eV}\, a_0^{-2}] \tag{57}$$

The $\hbar/e$ factor converts rad s^{-1} to eV.

2.8.3 Shell-Averaged Mean-Square Radius

$$\langle r^2 \rangle_n = \frac{1}{n^2} \sum_{l=0}^{n-1} (2l+1) \frac{n^2}{2Z^2} \left[5n^2 + 1 - 3l(l+1)\right] a_0^2 \tag{58}$$

This expression scales as n^4/Z^2 and is *not* replaced by an approximate constant.

2.8.4 Strong-Collision Constant

Values from Ferri, Peyrusse & Calisti [1], Table 1:

$$C_n = \begin{cases} 1.50 & n \leq 2 \\ 1.00 & n = 3 \\ 0.75 & n = 4 \\ 0.50 & n = 5 \\ 0.40 & n \geq 6 \end{cases} \tag{59}$$

2.8.5 Dynamical GBK Factor

$$G(\Delta E) = \frac{1}{2} E_1(y(\Delta E)), \qquad y(\Delta E) = \left(\frac{n^2}{2Z}\right)^2 \frac{\Delta E^2 + E_c^2}{\mathrm{Ry}_\infty T_e} \tag{60}$$

where $\Delta E = E - E_{pk}$ [eV] is the detuning from the transition center, $E_c = \hbar\omega_c$ [eV] is the cutoff energy, $\mathrm{Ry}_\infty \approx 13.606\,\mathrm{eV}$ is the Rydberg energy ($e^2/2a_0$, the ionization energy of hydrogen), and $E_1(x)$ is the first-order exponential integral function defined for $x > 0$ by:

$$E_1(x) = \int_x^\infty \frac{e^{-t}}{t}\, dt \tag{61}$$

Cutoff frequency. The collision integral diverges logarithmically at large impact parameters unless truncated at a maximum $\rho_{\max}$, or equivalently at a lower cutoff $\omega_c = v_{\mathrm{th}}/\rho_{\max}$ [8]. StarkZee follows the formulation of Ferri, Peyrusse & Calisti [1]:

$$\omega_c = \max\left(\omega_p,\ \omega_e,\ \omega_L,\ \omega_{\alpha\alpha'}\right) \tag{62}$$

- $\omega_p = \sqrt{N_e e^2/(\varepsilon_0 m_e)}$ — **electron plasma frequency**: Debye screening suppresses interactions at $\rho > \lambda_D = v_{\mathrm{th}}/\omega_p$.
- $\omega_e = v_{\mathrm{th}}/r_e$ — **configuration-change frequency**: the impact approximation requires each collision to be completed before the surrounding electron configuration changes appreciably. At high density or low T_e this can be shorter than the Debye timescale (see Section 2.8.6).
- $\omega_L = eB/m_e$ — **electron Larmor frequency**: in $\vec{B}$, electrons follow helical trajectories, reducing the effective interaction duration.
- $\omega_{\alpha\alpha'}$ — **emitter level-splitting frequency**: zero for hydrogen (degenerate l-subshells).

The largest frequency dominates, imposing the tightest bound on $\rho_{\max}$.

Frequency-dependent vs. resonance-center width. When `frequency_dependent_width=True` (default), $\gamma_e(\Delta E)$ is evaluated at the actual detuning of each transition component using the full $E_1(y(\Delta E))$ function. Setting this flag to `False` fixes the width at the line-center value $\gamma_e(0)$, which reduces computation time at the cost of accuracy in the far wings.

Application in the profile. Each Stark-dressed transition component $(i \to j, q)$ is broadened by a Lorentzian of half-width γ_e:

$$L_{ij}(E) = \frac{\gamma_e/\pi}{(E - E_{ij})^2 + \gamma_e^2} \tag{63}$$

and the contribution is weighted by the component intensity $|d_q(i,j)|^2$ and the microfield quadrature weight.

2.8.6 ZEST electron broadening variant (`electron_model='zest'`)

Activating `electron_model='zest'` in `LineProfile.compute_profile` switches to the electron broadening model of the ZEST code [17]. The formula has the same structure as the default GBK expression but replaces the full shell-averaged $\langle r^2 \rangle_n$ with the *intra-shell* mean-square radius:

$$\gamma_e^{\text{ZEST}}(\Delta E) = W_{\text{pref}} \, \langle r^2_{\text{intra}} \rangle_n \, [G_n + G_{\text{ZEST}}(\Delta E)] \tag{64}$$

Physical justification — $\Delta n = 0$ interaction channels. The ZEST paper (Section 2.1 of [17]) adopts two approximations that together restrict the electron broadening operator to within-shell matrix elements:

1. **No-quenching approximation**: perturbers may not induce transitions between the lower and upper manifolds participating in the radiator emission process.
2. $\Delta n = 0$ **interaction channels**: "only states belonging to the same Layzer complex [same principal quantum number n] may be mixed by Stark and/or Zeeman effects."

A Layzer complex is the set of all n^2 states sharing principal quantum number n. These two restrictions reduce the formal sum over all intermediate states in the broadening operator to within-shell dipole matrix elements only, directly yielding $\langle r^2_{\text{intra}} \rangle_n$ in place of the full $\langle r^2 \rangle_n$. The restriction is also automatic from the GBK dynamical factor: an inter-shell detuning (e.g. the $n = 3 \to n = 2$ gap of ≈ 1.89 eV at the conditions of Fig. 2) drives $G(\Delta\omega_{\text{inter}}) \to 0$ exponentially, so inter-shell contributions vanish regardless of the explicit approximation.

Intra-shell squared radius. Starting from the intra-shell radial element $\langle n, l | r | n, l \pm 1 \rangle = \frac{3n}{2Z}\sqrt{n^2 - (l \pm 1)^2}$ and angular weight factors $C(l, l+1) = (l+1)/(2l+1)$, $C(l, l-1) = l/(2l+1)$, the per-l intra-shell sum is

$$r^2_{\text{intra},l} = \frac{9n^2}{4Z^2}\left(n^2 - l(l+1) - 1\right) \quad [a_0^2] \tag{65}$$

This formula holds for all $l \in [0, n-1]$, including the boundary cases $l = 0$ (no $l-1$ neighbor) and $l = n-1$ (no $l+1$ neighbor). Averaging over the n^2 spatial states with $(2l+1)$ weight yields the exact closed form:

$$\langle r^2_{\text{intra}} \rangle_n = \frac{9n^2(n^2-1)}{8Z^2} \quad [a_0^2] \tag{66}$$

Numerical values for H ($Z = 1$): $n = 2$: $13.5\,a_0^2$; $n = 3$: $81\,a_0^2$; $n = 4$: $270\,a_0^2$. These are smaller than the full shell averages (33, 153, $468\,a_0^2$) by factors of 0.41, 0.53, and 0.58, respectively.

Maximum wave-number cutoff κ_m. The PPPB model fixes the minimum impact parameter as $\rho_{\min} = n^2 a_0/(2Z)$ (Bohr orbit) and uses a combined cutoff $\omega_c = \max(\omega_p, \omega_e, \omega_L)$. The ZEST convention instead defines a temperature-dependent wave-number upper limit

$$\kappa_m = \min\left(\frac{Z}{n^2 a_0}, \frac{Z\sqrt{2m_e k_B T_e}}{\hbar n^2}\right) \tag{67}$$

The geometric branch $Z/(n^2 a_0)$ dominates above $T_e \approx \mathrm{Ry}_\infty \approx 13.6\,\mathrm{eV}$; below that the thermal de Broglie branch takes over. The only cutoff frequency retained in all ZEST G-functions is the electron plasma frequency ω_p (no ω_e or ω_L terms), so B does not enter the ZEST broadening directly.

G-function choices (`electron_model` selector). Three alternatives are available for the dynamical factor $G(\Delta\omega)$ in Eq. (55), all sharing $\langle r^2_{\text{intra}}\rangle_n$, the same $G_n = C_n$ strong-collision constants, and κ_m as defined above.

`zest` / `zest-gbk` (default ZEST). Closed-form exponential integral with a κ_m-derived argument:

$$G_{\text{GBK}}(\Delta\omega) = \tfrac{1}{2}\,E_1\!\left(\frac{\Delta\omega^2 + \omega_p^2}{2\,x^2\,\omega_p^2}\right), \qquad x = \kappa_m \lambda_D \tag{68}$$

This has the same E_1 structure as the PPPB formula but with $x = \kappa_m\lambda_D$ playing the role of $\rho_{\min}/\lambda_D$, and only ω_p in the additive term.

`zest-lee`. Analytical *min*-blend of the impact ($\Delta\omega \to 0$) and wing ($\Delta\omega \to \infty$) limits:

$$\begin{aligned} G_0 &= \tfrac{1}{2}\left[\ln(1+x^2) - \frac{x^2}{1+x^2}\right], \\ G_\infty(\Delta\omega) &= \tfrac{1}{2}\,E_1\!\left(\frac{\Delta\omega^2}{2\,x^2\,\omega_p^2}\right), \\ G_{\text{Lee}}(\Delta\omega) &= \min\big(G_0,\, G_\infty(\Delta\omega)\big) \end{aligned} \tag{69}$$

G_0 (the line-center plateau) provides the plasma-cutoff floor; the E_1 wing term contains only $\Delta\omega^2$ (no $+\omega_p^2$) because the regularization is already captured by G_0. At large detuning the two forms agree.

`zest-dufty`. Full random-phase-approximation (RPA) numerical integration accounting for Landau damping:

$$G_{\text{Dufty}}(\Delta\omega) = \int_{\kappa_{\min}}^{\kappa_m} \frac{e^{-x^2}}{\kappa\,|\varepsilon(\kappa,\Delta\omega)|^2}\,d\kappa, \qquad x = \frac{|\Delta\omega|}{\kappa\,v_{\text{th}}}, \quad v_{\text{th}} = \sqrt{\frac{2k_B T_e}{m_e}} \tag{70}$$

where the longitudinal RPA dielectric function is

$$\varepsilon(\kappa,\Delta\omega) = 1 + \frac{1}{(\kappa\lambda_D)^2}\left[1 - 2x\,\mathcal{D}(x) + i\sqrt{\pi}\,x\,e^{-x^2}\right] \tag{71}$$

and $\mathcal{D}(x) = e^{-x^2}\int_0^x e^{t^2}\,dt$ is the Dawson function. The lower limit $\kappa_{\min} = |\Delta\omega|/(10\,v_{\rm th})$ removes the static ($\kappa \to 0$) divergence. This is the most accurate of the three models but evaluates one numerical quadrature per frequency point.

In practice the three G-functions give nearly identical line cores; differences appear in the far wings where Landau damping becomes significant.

2.9 Profile Accumulation

Each Stark-dressed transition $(p \to k, q)$ is broadened by a Lorentzian of HWHM $\gamma_e(\Delta E)$:

$$L_{pk}(\Delta E) = \frac{1}{\pi}\,\frac{\gamma_e(\Delta E)}{\Delta E^2 + \gamma_e(\Delta E)^2} \tag{72}$$

The three polarization profiles are accumulated by vectorized summation over all microfield quadrature points and all SDTs:

$$P_q(E) = \sum_i W_{F,i} \sum_j W_{\mu,j} \sum_{p,k} S_q(p,k)\, L_{pk}(E - E_{pk}) \tag{73}$$

where $q{=}0 \to P_\pi$, $q{=}{-1} \to P_{\sigma^+}$ (blue), $q{=}{+1} \to P_{\sigma^-}$ (red). This direct summation realizes Eq. (10) in practice.

Frequency-dependent vs. resonance-center width. When `frequency_dependent_width=True` (default), $\gamma_e(\Delta E)$ is evaluated at the actual detuning of each (E, E_{pk}) pair using the full $E_1(y(\Delta E))$ function. Setting this flag to `False` fixes the width at the line-center value $\gamma_e(0)$, reducing computation time at the cost of accuracy in the far wings.

Application in the profile. Each component is weighted by its SDT intensity $S_q(p,k)$ and the microfield quadrature weight. The observed intensity at angle α to $\vec{B}$ is then:

$$I(\omega,\alpha) = P_\pi(\omega)\sin^2\alpha + \tfrac{1}{2}\big(P_{\sigma^+}(\omega) + P_{\sigma^-}(\omega)\big)(1+\cos^2\alpha) \tag{74}$$

2.10 Thermal Doppler Broadening

Thermal motion of the radiating ions causes each photon frequency to be Doppler-shifted by $\delta\omega = \omega_0\, v_z/c$. Averaging over a Maxwell–Boltzmann velocity distribution yields a Gaussian line shape with $1/e$ half-width

$$\Delta\omega_D = \omega_0\sqrt{\frac{2k_B T_i}{m_{\rm ion}c^2}} \qquad \Longleftrightarrow \qquad \Delta E_D = E_0\sqrt{\frac{2T_i}{m_{\rm ion}c^2/e}} \tag{75}$$

where T_i is the ion temperature, $m_{\rm ion}$ the emitter mass, and E_0 the transition energy in eV. For H Balmer-α at $T_i = 5\,$eV, this gives $\Delta E_D \approx 0.062\,$meV — comparable to the Zeeman splitting at 1 T and negligible relative to the Stark width at $N_e \sim 10^{23}\,\mathrm{m}^{-3}$.

Important: Doppler broadening is *not* applied automatically. `calculate_static_profile()` returns the purely Stark-Zeeman-broadened profile (ion quasi-static + electron impact only). Doppler broadening must be applied *after* as a post-processing step via `convolutions.py`:

- `apply_doppler_broadening(wavelengths_nm, profile, Ti_ev, species='H')` — convolves the profile with a Gaussian kernel of the thermal Doppler width.

- `apply_instrument_broadening(wavelengths_nm, profile, fwhm_nm)` — convolves with a Gaussian instrumental slit function.

Both use FFT convolution (`scipy.fft`) for efficiency. Both operate on a *uniform wavelength grid*; convert energy grids to wavelength before calling. The convolution preserves the total integrated intensity.

The Gaussian kernel applied to the wavelength grid is:

$$K_D(\lambda) = \exp\left(-\frac{(\lambda-\lambda_0)^2}{\Delta\lambda_D^2}\right) \tag{76}$$

Convolution is performed via the FFT theorem on a wavelength grid edge-padded by $N/2$ on each side to avoid circular-wrap artifacts:

$$P_{q,\mathrm{conv}}(\lambda) = \mathcal{F}^{-1}\big[\mathcal{F}(P_{q,\mathrm{pad}}) \cdot \mathcal{F}\big(\mathrm{fftshift}(K_{D,\mathrm{pad}})\big)\big] \tag{77}$$

The complete broadening pipeline is therefore:

$$I_{\mathrm{obs}}(\lambda) = \Big[I_{\mathrm{SZ}}(\lambda) * G_D(\lambda)\Big] * G_{\mathrm{inst}}(\lambda) \tag{78}$$

where I_{SZ} is the Stark-Zeeman profile from `static_profile.py` (via `line_profile.py`), G_D is the Doppler Gaussian, and G_{inst} the instrumental Gaussian. Both convolutions preserve the total integrated intensity.

2.11 Frequency Fluctuation Model (FFM)

Dynamic ion motion causes the static microfield to fluctuate over time. In the FFM [6], this is modeled as a stationary Markovian process that mixes the Stark-dressed transition components at rate ν_i.

The Stark-Zeeman Hamiltonian $H = H_A + V_E$ is still diagonalized at each field configuration (F, μ), producing dressed-state frequencies ω_k and dipole weights $|d_k|^2$ — the Stark-Dressed Transitions (SDTs). This step is purely static: ω_k and $|d_k|^2$ depend only on the instantaneous ion field. Fast electron collisions add a homogeneous Lorentzian half-width γ_k to each SDT (the GBK width from Section 2.8), acting on a timescale short enough that the ion configuration does not change during a single collision. Ion dynamics enter exclusively through ν_i, estimated as the inverse time for an ion to cross the mean ion spacing at its thermal speed:

$$\nu_i = \frac{v_{th}}{r_i}, \qquad v_{th} = \sqrt{\frac{2k_BT_i}{m_i}}, \qquad r_i = \left(\frac{3}{4\pi N_i}\right)^{1/3} \tag{79}$$

The dynamic line profile is then given by the Sherman-Morrison form:

$$I(\omega) = \frac{r^2}{\pi}\mathrm{Re}\frac{S(\omega)}{1-\nu_i S(\omega)} \tag{80}$$

with the static propagator sum:

$$S(\omega) = \sum_k \frac{p_k}{\nu_i + \gamma_k + i(\omega-\omega_k)} \tag{81}$$

where $p_k = |d_k|^2/r^2$ are the normalized SDT weights and $r^2 = \sum_k |d_k|^2$. The limit $\nu_i \to 0$ recovers the static profile; $\nu_i \to \infty$ gives a single Lorentzian (motional narrowing).

3 Physical Features and Validation

3.1 Quadratic Zeeman polarization wings

At $B \geq 500\,\mathrm{T}$ the diagonal QZ shifts within $n = 3$ differ by up to 5 meV, separating the 3p($m_l = \pm 1$)→2s transitions (which carry ≈ 21% of the Hα oscillator strength and exhibit the *largest* n=3 QZ shift, +8.87 meV at $B = 1000\,\mathrm{T}$) from the dominant 3d→2p cluster by ≈ 7 meV. The resulting "wings" appear only in the `quadratic_zeeman=True` profiles and are validated in `test_12_halpha_qz_wings.py`.

3.2 $\pm 2\mu_B B$ Stark-Zeeman satellite features

A transverse microfield F_x couples adjacent-m_l states within the same principal shell via $\Delta l = \pm 1$, $\Delta m_l = \pm 1$ matrix elements. The upper eigenstate near $E_{0,n} + 2\mu_B B$ (dominated by $|nd,\, m_l{=}2\rangle$) acquires a small $|np,\, m_l{=}1\rangle$ admixture β, producing a σ^+ transition to the zero-Zeeman lower state at photon energy $E_0 + 2\mu_B B$.

Hβ (n=4→2). The satellite is a *distinct peak* at +117 meV (≈ 2% of the main σ^+ intensity), because $n = 4$ has both $|4d,\, m_l{=}2\rangle$ and $|4f,\, m_l{=}2\rangle$ *degenerate* at $+2\mu_B B$, greatly amplifying Stark mixing. The satellite is visible in the Balmer-series profile at ≈ 465 nm and ≈ 509 nm for $B = 1000\,\mathrm{T}$.

Hα (n=3→2). Only $|3d,\, m_l{=}2\rangle$ sits at $+2\mu_B B$ (no $l = 3$ sub-states exist for $n = 3$). The satellite amplitude (≈ 0.07% of main peak) is comparable to the Lorentzian tail of the σ^+ main peak at that position, so no distinct local maximum appears at 618 nm / 700 nm. This result is converged with respect to the microfield grid size (`num_f` = 30, 60, 100 give identical profiles) and is validated in `test_13_stark_zeeman_satellites.py`.

***B*-dependence and observability.** The mixing coefficient $\beta = F_x \langle np|r|nd\rangle / (\mu_B B)$ scales as B^{-1}, so β is *larger* at lower B. However, the satellite position $2\mu_B B$ also shrinks with B. At $B = 100\,\mathrm{T}$, $2\mu_B B = 11.6\,\mathrm{meV}$ falls inside the Stark-broadened σ^+ cluster (~ 10 meV width for $n = 4$, $N_e = 10^{23}\,\mathrm{m}^{-3}$). The satellite merges into the cluster tail: the peak/valley contrast in the satellite sub-window is ≈ 1.000 (indistinguishable), versus ≈ 2.26 at $B = 1000\,\mathrm{T}$ where the satellite is clearly separated. *High B is therefore required for the satellite to appear as a resolved, separated peak.*

4 Physical and Numerical Simplifications

To achieve absolute self-sufficiency, maintainability, and high execution speeds, the following simplifications were incorporated:

1. **Analytical Hydrogenic Basis**: Instead of calling external commercial databases or complex atomic structure codes (e.g., Cowan or FAC), SZLS implements exact analytical hydrogenic radial wavefunctions [3]:

$$R_{nl}(r) = \sqrt{\left(\frac{2Z}{n}\right)^3 \frac{(n-l-1)!}{2n(n+l)!}}\, e^{-Zr/n} \left(\frac{2Zr}{n}\right)^l L_{n-l-1}^{2l+1}\left(\frac{2Zr}{n}\right) \tag{82}$$

This provides self-contained matrix elements for r and r^2 calculations.

2. **Resonance-center impact approximation (optional)**: By default (`frequency_dependent_width=True`) the GBK electron width $\gamma_e(\Delta E)$ is evaluated at the actual detuning of each transition component. Setting `frequency_dependent_width=False` fixes the width at its line-center value $\gamma_e(0)$, avoiding repeated evaluations of $E_1(y)$ at every wavelength point. This is a useful approximation when the profile wings are not of primary interest.
3. **Gauss-Legendre Angle Quadrature**: Integrating over the field orientation is performed using a N_μ-point Gauss-Legendre quadrature over $\mu = \cos\theta \in [0, 1]$; see Section 2.7 for the full discretization scheme.

5 Built-in Reference Models

The `starkzee.models` package provides five independent lineshape models that share a common call signature and serve as benchmarks against StarkZee's fully coupled solver. They are grouped into tabulated models (reading precomputed databases) and analytical models.

5.1 Tabulated Models

5.1.1 *Stehlé (MMM) —* `stehle`

The Stehlé model reads precomputed Stark lineshapes from the Model Microfield Method (MMM) database [14] for hydrogen Balmer and Paschen transitions over a grid of N_e and T_e values. The tables contain the *unmagnetized* ($B = 0$) Stark + fine-structure profile as a function of a reduced detuning $\Delta\omega/F_0$ (Holtsmark normal field units). The pipeline is:

1. Bilinear interpolation over the (N_e, T_e) table grid (with a Fortran-style three-point hyperbolic interpolator for the detuning axis).
2. Thermal Doppler broadening via FFT convolution.
3. Normal Zeeman splitting applied *after* the table: the whole Stark+Doppler profile is rigidly copied to $\omega \pm \omega_L$ (Larmor frequency) with angle-dependent π/σ weights:

$$I(\theta) = \sin^2\theta\, I_\pi + \tfrac{1+\cos^2\theta}{2}\,(I_{\sigma+} + I_{\sigma-}) \tag{83}$$

The Stark and Zeeman effects are therefore treated as **separable**. This is valid only when the Zeeman splitting $\mu_B B$ is much smaller than the Stark width. Stehlé covers arbitrary (n_u, n_l) and species with no restriction on B-field value (it is simply ignored in the table lookup).

5.1.2 *Rosato —* `rosato`

The Rosato database [15] solves the Stark-Zeeman problem *jointly* and tabulates the result with B as an explicit table axis ($B \in \{0, 1, 2, 2.5, 3, 5\}$ T). The pipeline is:

1. Interpolation over the (N_e, T_e, B) grid. The B-interpolation is *scaled*: before blending two bracketing profiles at B_0 and B_1, each profile's detuning axis is stretched by B_{node}/B. This exploits the fact that Zeeman splitting scales linearly with B, aligning the σ components before interpolating.

2. Angle dependence from *two real tables* (parallel and perpendicular to $\vec{B}$), blended as $I = I_{\parallel} \cos^2\theta + I_{\perp} \sin^2\theta$. Both tables carry the correct π/σ lineshape and width — the angle enters as a physical interpolation, not just an amplitude weight.
3. Thermal Doppler broadening via FFT convolution.

Rosato is restricted to deuterium Balmer lines and the table coverage $N_e \in [10^{13}, 10^{16}]$ cm^{-3}, $T_e \in [0.316, 31.6]$ eV, $B \leq 5$ T.

5.2 Analytical Models

5.2.1 *Lomanowski — `lomanowski`*

Polynomial fits for the Stark FWHM from Lomanowski *et al.* [16]:

$$\Delta\lambda_S = c_{n_u n_l}\, N_e^{a_{n_u n_l}}\, T_e^{-b_{n_u n_l}} \quad [\mathrm{nm}] \tag{84}$$

Coefficients (a, b, c) are tabulated for H/D Balmer and Paschen lines up to $n_u = 9$. A Voigt profile combines this Stark Lorentzian with a Doppler Gaussian. No magnetic field treatment.

5.2.2 *Parameterized Stehlé — `stehle_param`*

An analytical parametric fit to the Stehlé tables, providing a fast closed-form approximation to the field-free Stark profile without reading the NetCDF database.

5.2.3 *Voigt — `voigt`*

A pseudo-Voigt profile using the Griem α_{12} Stark half-width [7] combined with thermal Doppler broadening. Serves as a quick estimate when only order-of-magnitude Stark width accuracy is required.

5.3 Comparison with StarkZee

Feature	**StarkZee**	**Stehlé**	**Rosato**
Magnetic field	Full simultaneous diagonalization of $H = H_A + V_E$	$B = 0$ in tables; added as rigid triplet after	B is a table axis
Fine structure	Yes — spin-orbit, MV, Darwin	No (degenerate hydrogenic)	In tables
Electron broadening	GBK semi-classical, frequency-dependent	Unified theory (more rigorous far wings)	In tables
Ion dynamics	Quasi-static + optional FFM	Static + some ion-dynamics treatment	In tables
Microfield	Hooper screened	Holtsmark/Hooper variant	Holtsmark variant
B-treatment	Intrinsic to Hamiltonian	External convolution	B-scaled interpolation

The most important physical distinction at $B \neq 0$: StarkZee diagonalizes $H = H_A + V_E$ *simultaneously* for all field strengths. When $\mu_B B \sim 3n\, e\, a_0\, F$ (Zeeman and Stark splitting comparable) the eigenstates are genuine Stark-Zeeman hybrids — neither pure Zeeman nor pure Stark states. No post-processing convolution can reproduce this mixing. The separable approximation (Stehlé) and scaled-interpolation approach (Rosato) both become inaccurate in this regime.

6 Code Structure and Modular Architecture

The **StarkZee** package (`starkzee/`) is organized as a modular Python library:

- `starkzee/line_profile.py`: High-level `LineProfile` class. Manages spectral grids, unit conversions, Doppler broadening, and exposes all results as attributes. Entry point for most users.
- `starkzee/static_profile.py`: Core solver. `calculate_static_profile()` runs the microfield quadrature loop and accumulates the three polarization profiles P_π, P_{σ^+}, P_{σ^-}.
- `starkzee/radiator.py`: Constructs H_A — the atomic Hamiltonian including unperturbed energy, spin-orbit, fine-structure, and linear and quadratic Zeeman terms — and builds the dipole operator matrices using analytical angular matrix elements. H_A is *not* diagonalized here; the full $H = H_A + V_E$ is diagonalized in `solve_starkzee`.
- `starkzee/ffm.py`: Implements dynamic ion-field fluctuations via the Frequency Fluctuation Model, supporting both an analytical Sherman-Morrison solver and a full matrix inversion.
- `starkzee/microfield.py`: Generates screened Hooper and unscreened Holtsmark microfield distributions; supports multi-species Debye lengths and custom tabulated distributions.
- `starkzee/broadening.py`: GBK electron-impact half-width $\gamma_e(\Delta E)$; computes plasma, Larmor, and configuration-change cutoff frequencies.
- `starkzee/convolutions.py`: FFT-based post-processing convolutions operating on uniform wavelength grids:

$$I_{\text{conv}}(\lambda) = \mathcal{F}^{-1}\big\{\mathcal{F}[I(\lambda)] \cdot \mathcal{F}[K_{\text{Doppler}}(\lambda)] \cdot \mathcal{F}[K_{\text{slit}}(\lambda)]\big\} \tag{85}$$

- `starkzee/models/`: Comparison lineshape models (voigt, lomanowski, stehle_param, stehle, rosato) sharing a common signature, centered on NIST air wavelengths for each transition [13].
- `starkzee/atomic_loader.py`: Placeholder for future multi-electron species support.

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A Algorithmic Overview of the Calculation Pipeline

When a user invokes `LineProfile.compute_profile(grid, grid_type)`, the `StarkZee` package performs a series of nested calculations to solve the coupled Stark-Zeeman line-shape problem. The numerical delegation chain is structured as follows:

1. `LineProfile.compute_profile()`: Manages the spectral coordinate axes, performs grid unit conversions, and forwards the arguments to the profile calculations.
2. `static_profile.calculate_static_profile()`: Generates the plasma microfield grid and weights, computes the Gauss-Legendre quadrature points for the microfield orientation angle θ, and runs the main double-integration loop.
3. `solve_starkzee()`: Assembles the full electron-radiator Hamiltonian $H = H_A + V_E$ (Eq. 25) and diagonalizes it as a single matrix with `numpy.linalg.eigh`. V_E is *not* applied perturbatively.
4. `radiator.build_hamiltonian()`: Builds H_A — the field-free plus magnetic Hamiltonian (unperturbed energy, spin-orbit, fine structure, linear and quadratic Zeeman) in the uncoupled $|n, l, m_l, m_s\rangle$ basis.
5. `static_profile._stark_templates()`: Returns the field-independent Stark coupling matrices M_z and M_x [eV/(V/m)], cached per (n, Z), so that $V_E = F_z M_z + F_x M_x$ is formed at each quadrature point without rebuilding the matrix loop.
6. **Transition Dipole Rotation**: Rotates the transition dipole operator matrices D_q to the mixed eigenstate bases.
7. **Electron-Impact Broadening**: Evaluates the frequency-dependent electron Stark half-width $\gamma_e(\Delta E)$ via the Griem-Baranger-Kolb (GBK) model and accumulates Lorentzian profiles.
8. `convolutions.apply_doppler_broadening()`: Performs a post-processing Fast Fourier Transform (FFT) convolution to apply thermal Doppler broadening.

B Physical Approximations and Domains of Validity

Approximation	Mathematical form	Domain of validity	Failure mode
Within-Shell Isolation ($\Delta n = 0$)	$V_E = \mathrm{diag}_n(V_E)$, no $n \to n\pm1$ coupling	Stark shift $\ll$ shell spacing $2Z^2\mathrm{Ry}/n^3$	Quadratic Stark; Inglis-Teller merging at $N_e \gtrsim 10^{24}\,\mathrm{m}^{-3}$.
Quasi-Static Ion Microfields	$\vec{F}_{\mathrm{ion}} = \mathrm{const}$	Fluctuation rate $\nu_i \ll$ Stark width ΔE_S	Ion dynamics / motional narrowing; use FFM.
Semi-Classical Electron Impact	Lorentzian with GBK width γ_e	$\lambda_{dB} \ll$ impact parameter ρ	Quantum close-coupling needed for cold dense plasmas.
Dipole Approximation	$V_{\mathrm{rad}} \propto \vec{r} \cdot \vec{E}$	$\lambda \gg \langle r \rangle_n \sim n^2 a_0/Z$	Quadrupole/octupole for high-n Rydberg states.

Table 1: Key physical approximations, their domains of validity, and failure modes.

C Physics Explanation of Use Cases and Operational Limits

C.1 The High-Magnetic-Field Regime

In laboratory magnetic confinement fusion devices (tokamaks, stellarators), the magnetic field reaches $B \approx 1$–$10\,\mathrm{T}$:

- The Zeeman splitting $\Delta E_Z \approx 0.1\,\mathrm{meV}$ is comparable to the fine-structure splitting of lower shells, requiring the full coupled Hamiltonian diagonalization.
- Typical microfield strengths $F \sim 10^5$–$10^6\,\mathrm{V/m}$ represent a weak-Stark regime where Stark broadening acts as a symmetric perturbation on the Zeeman triplet structure.

For astrophysical compact objects (magnetized white dwarfs with $B \sim 10^3$–$10^5\,\mathrm{T}$, neutron stars with $B \sim 10^8\,\mathrm{T}$):

- The quadratic Zeeman term $H_Z^{(2)} \propto B^2$ dominates, causing significant blue-shifting and highly asymmetric splitting. The `quadratic_zeeman=True` flag enables exact numerical treatment of the $\Delta l = \pm 2$ coupling (see Section 2.2).
- `StarkZee`'s exact numerical computation of $\langle n, l_1 | r^2 | n, l_2 \rangle$ avoids the geometric-mean overestimation of up to 41% for $n = 5$.

C.2 Density Limits

- **Low density** ($N_e \lesssim 10^{18}\,\mathrm{m}^{-3}$): The profile is dominated by thermal Doppler broadening; microfield weights converge to a delta function at $F = 0$.
- **High density** ($N_e \gtrsim 10^{24}\,\mathrm{m}^{-3}$): The inter-particle spacing r_e approaches the atomic radius $\langle r \rangle_n$. The static and $\Delta n = 0$ approximations break down as the Stark shift exceeds the Rydberg shell spacing (Inglis-Teller limit).